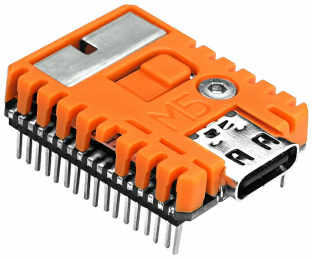
Stamp-S3A PIN1.27

SKU:S007-PIN127-V033

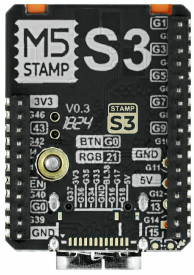


























Description

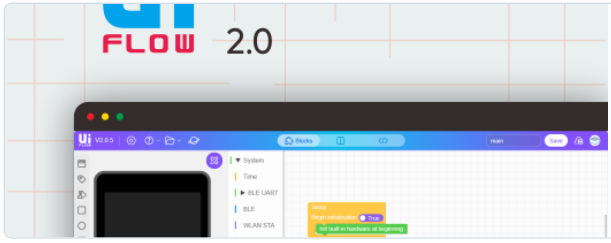
Stamp-S3A PIN1.27 is a highly-integrated embedded module based on the Espressif ESP32-S3FN8. It features a 240 MHz Xtensa® 32-bit LX7 dual-core processor, integrates 8 MB Flash, a programmable RGB LED, and a user button. The deeply optimized 3D antenna design delivers stronger wireless performance. Twenty-three GPIOs are broken out and a 1.27 mm-pitch SMT pin header is pre-soldered, providing an even smaller footprint for high-density PCB integration and enabling developers to build compact applications quickly.

Tutorial



Arduino IDE

This tutorial shows you how to program and control the Stamp-S3A PIN1.27 with the Arduino IDE.



UiFlow2

This tutorial shows you how to control the Stamp-S3A PIN1.27 using the UiFlow2 graphical programming platform.

Features

- ESP32-S3FN8 (2.4 GHz Wi-Fi)
- 1.27 mm-pitch SMT interface
- 23 multifunctional GPIOs
- 8 MB Flash
- Rich I/O breakout supporting multiple application forms (SMT, DIP, fly-wire)
- Integrated programmable RGB LED and button
- Development Platform
  - UiFlow2
  - Arduino IDE
  - ESP-IDF
  - PlatformIO

Includes

- 1 x Stamp-S3A PIN1.27
- 1 x Pin Sticker

Applications

- Smart home
- Intelligent IoT devices
- Industrial control terminals
- Wearable devices

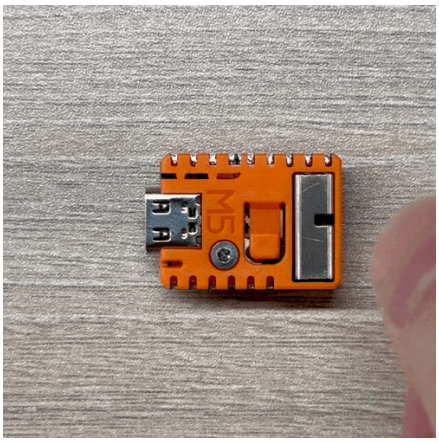
Specifications

Specification	Parameter
SoC	ESP32-S3FN8 @ Xtensa® 32-bit LX7 dual-core processor, clock frequency 240MHz
Flash	8MB
DC-DC	MUN3CAD01-SC
Wi-Fi	2.4 GHz Wi-Fi
Input Voltage	DC 5V
HMI	Programmable physical button × 1, programmable RGB LED (WS2812B-2020) × 1
Antenna Type	2.4 GHz 3D antenna
Module Resource I/F	Touch Sensor, SD/SDIO/MMC Host Ctrl, SPI, SDIO/SPI Slave Ctrl, EMAC, Motor PWM, LED PWM, UART, I2C, I2S, GPIO, Pulse Counter
IO Interfaces × 23	G0/G1/G2/G3/G4/G5/G6/G7/G8/G9/G10/G11/G12/G13/G14/G15/G39/G40/G41/G42/G43/G44/G46
IO Pitch	1.27 mm
LCD FPC Pitch	0.5 mm @ 12 P or 8 P
Operating Temp.	0 ~ 40 °C
Product Size	26.0 × 18.0 × 4.7 mm
Product Weight	3.1 g
Package Size	74.0 × 48.0 × 13.0 mm
Gross Weight	7.6 g

Learn

Download Mode

To enter download mode, hold the G0 button on the Stamp-S3A PIN1.27 before powering on, then release it after power is applied.

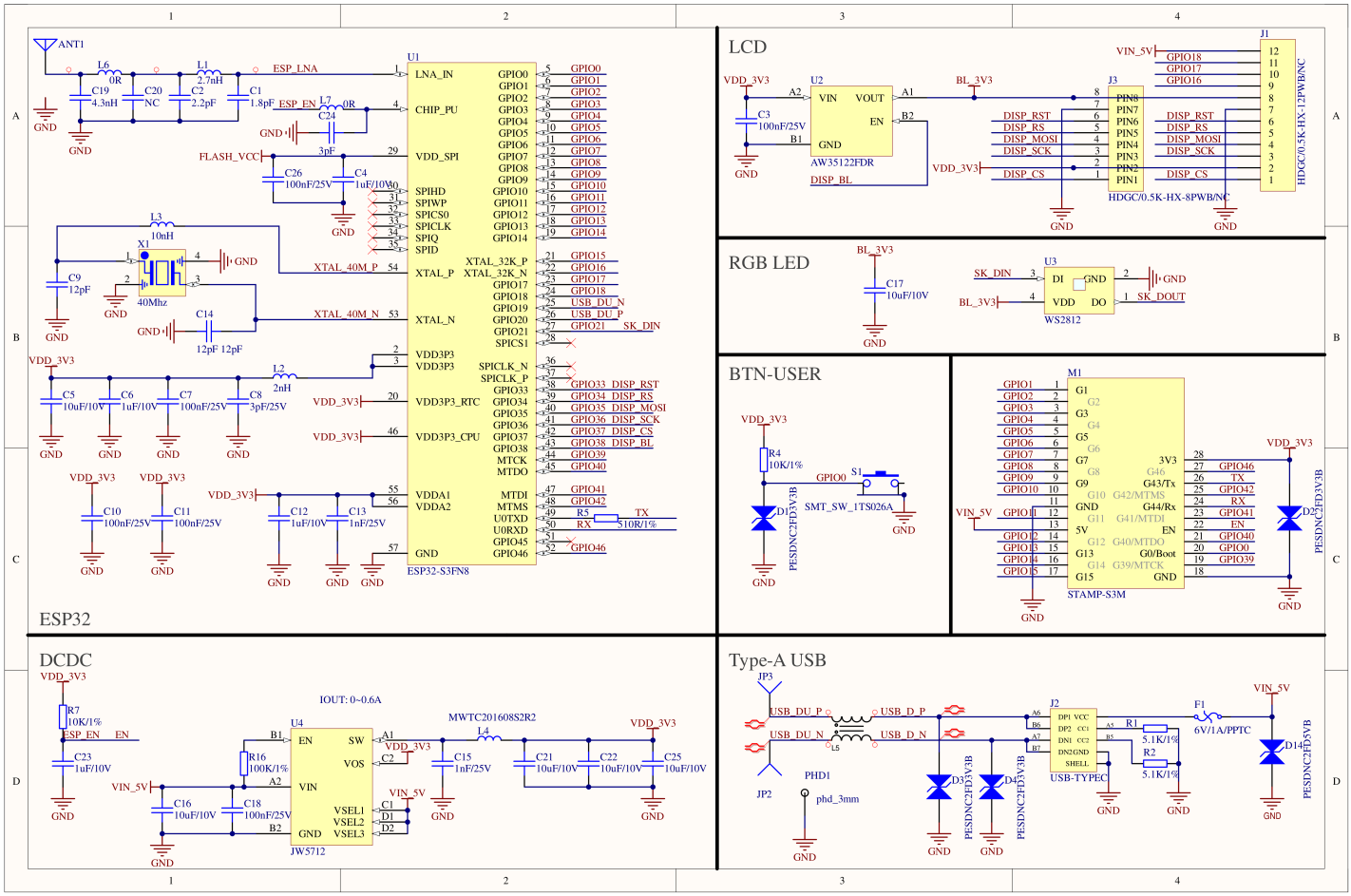


Additional Notes

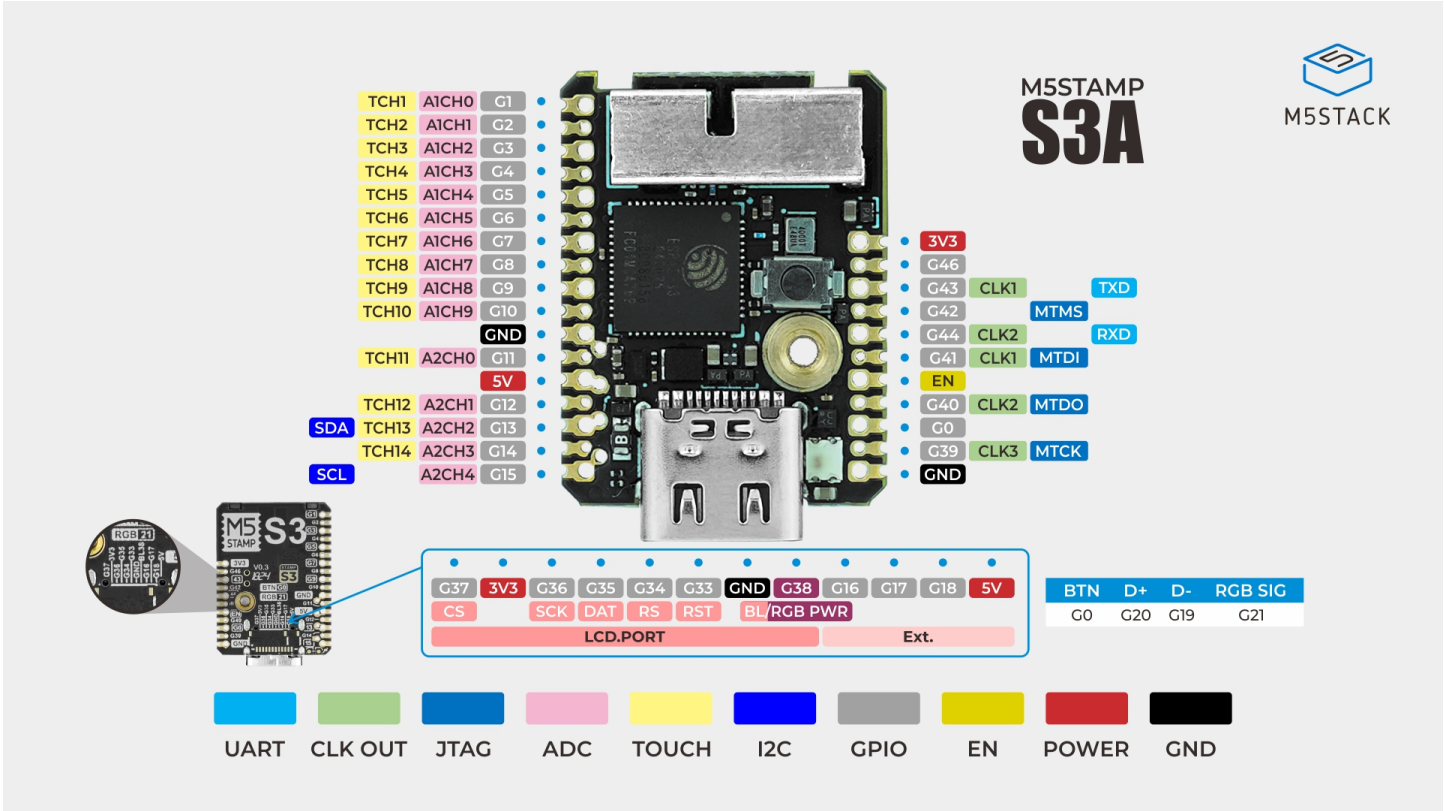
- ESP32-S3 pins G0 and G46 are strapping pins that determine the boot mode. By default, G0 is pulled-up and G46 is internally pulled-down, enabling the chip to boot from SPI Flash normally. Do not pull G46 high before the chip boots; otherwise, the chip will fail to start. Refer to the Strapping-Pin section of the ESP32-S3 datasheet for details.
- An SPI interface for a TFT display is reserved on the back of the module. Compatible FPC connector specifications are 8 PIN and 12 PIN.

Schematics

- Stamp-S3A PIN1.27 Schematics PDF

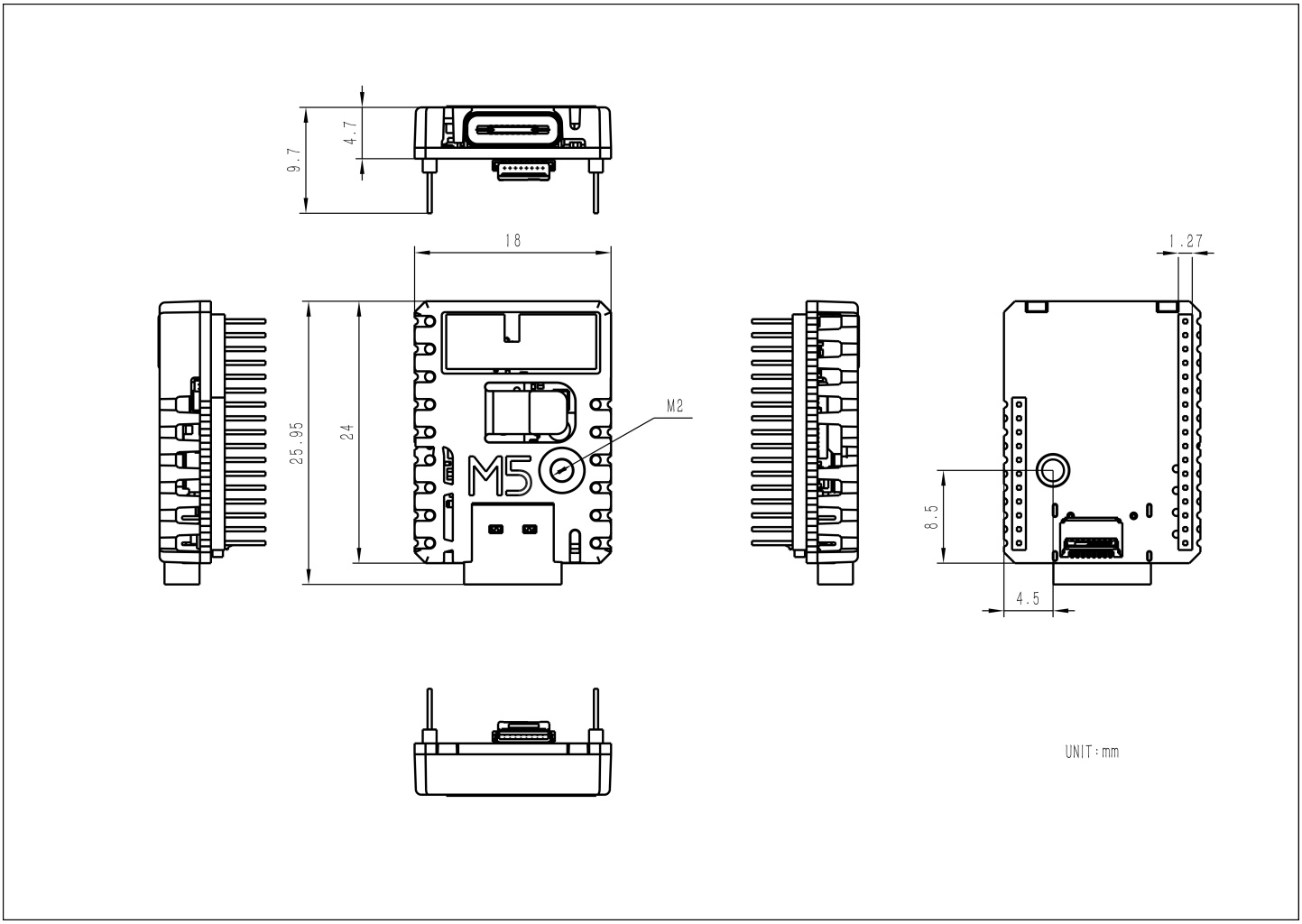

ESP32
LCD
RGB LED
BTN-USER
DCDC
Type-A USB
ESP32-S3FN8
AW35122FDR
WS2812
STAMP-S3M
JW5712
USB-TYPEC

PinMap


M5STAMP
S3A
M5STACK
TCH1 A1CH0 G1
TCH2 A1CH1 G2
TCH3 A1CH2 G3
TCH4 A1CH3 G4
TCH5 A1CH4 G5
TCH6 A1CH5 G6
TCH7 A1CH6 G7
TCH8 A1CH7 G8
TCH9 A1CH8 G9
TCH10 A1CH9 G10
GND
TCH11 A2CH0 G11
5V
TCH12 A2CH1 G12
SDA TCH13 A2CH2 G13
TCH14 A2CH3 G14
SCL A2CH4 G15
3V3
G46
G43 CLK1 TXD
G42 MTMS
G44 CLK2 RXD
G41 CLK1 MTDI
EN
G40 CLK2 MTDO
G0
G39 CLK3 MTCK
GND
G37 3V3 G36 G35 G34 G33 GND G38 G16 G17 G18 5V
CS SCK DAT RS RST BL/RGB PWR
LCD.PORT Ext.
BTN D+ D- RGB SIG
G0 G20 G19 G21
UART CLK OUT JTAG ADC TOUCH I2C GPIO EN POWER GND

Model Size

Stamp-S3A PIN1.27 Model Size PDF



PCB

- Stamp-S3A PIN1.27 KiCad Library

Structure

- Stamp-S3 KiCad 3D

Datasheets

- ESP32-S3
- MUN3CAD01-SC

Softwares

Arduino

- Stamp-S3A PIN1.27 Arduino Quick Start

UiFlow2

- Stamp-S3A PIN1.27 UiFlow2 Quick Start

Video

Product Comparison

Product Comparison Item		
	Stamp-S3A PIN1.27	**Stamp-S3 PIN1.27**
RGB LED Control Logic	RGB LED power is multiplexed with the reserved screen FPC bus backlight	Powered as soon as power is applied
Antenna Design	Optimized antenna design with better signal reception	Conventional antenna design
Module Boot Button	Improved tactile feel; 4.0 × 3.0 × 2.0 mm button	2.6 × 1.6 × 0.55 mm button
Power Consumption	Optimized for lower power consumption	Standard design

To compare information on the Stamp series products, you can visit the Product Selection Table, check the target products, and get the comparison results. The selection table covers key information such as core parameters and functional features, and supports comparison of multiple products simultaneously.